Class 12: RNASeq Galaxy

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Section 1: Proportion of G/G in a population

Downloaded a CSV file from Ensemble < https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39894595-39895595;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read this CSV file

```r
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")

head(mxl)
```

```
  Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                  NA19648 (F)                       A|A ALL, AMR, MXL      -
2                  NA19649 (M)                       G|G ALL, AMR, MXL      -
3                  NA19651 (F)                       A|A ALL, AMR, MXL      -
4                  NA19652 (M)                       G|G ALL, AMR, MXL      -
5                  NA19654 (F)                       G|G ALL, AMR, MXL      -
6                  NA19655 (M)                       A|G ALL, AMR, MXL      -
  Mother
1      -
2      -
3      -
4      -
```

```
5       -
6       -
```

Access column of mxl object containing genotype information, and feed this to `table()` function:

```
table(mxl$Genotype..forward.strand.)
```

```

A|A A|G G|A G|G 
 22  21  12   9 
```

```
round(table(mxl$Genotype..forward.strand.) / nrow(mxl) * 100, 2)
```

```

  A|A   A|G   G|A   G|G 
34.38 32.81 18.75 14.06 
```

Now let's look at a different population. I picked the GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Find proportion of G|G.

```
round(table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100, 2)
```

```

  A|A   A|G   G|A   G|G 
25.27 18.68 26.37 29.67 
```

This varient that is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Let's now dig into this further.

Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

> Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```r
# Read the file into R.
expr <- read.table("rs8067378_ENSG00000172057.6.txt")

# Preview the first 6 rows (samples) of expr.
head(expr)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

How many samples do we have?

```r
nrow(expr)
```

```
[1] 462
```

We have 462 samples in total.

Determining sample size per genotype:

```r
table(expr$geno)
```

```

A/A A/G G/G
108 233 121
```

The sample size for each genotype is as follows: 108 samples for A/A, 233 samples for A/G, and 121 samples for G/G.

Determining median expression per genotype:

Method 1: My preferred method is to use the dplyr package's `group_by()` (to group the samples by genotype) and `summarise()` functions (to calculate the median expression level for each group).

```
library(dplyr)
```

```
Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

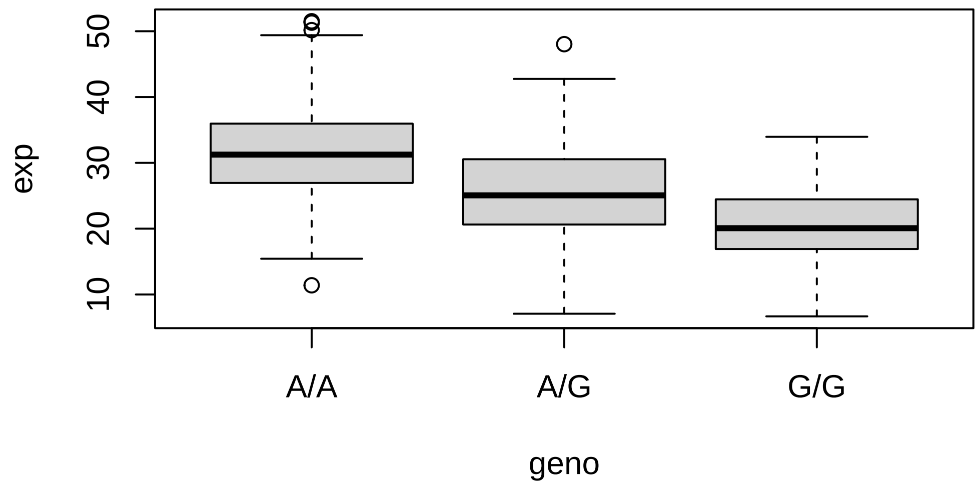
```
expr.median <- expr |>
  group_by(geno) |>
  summarise(median = median(exp))
expr.median
```

```
# A tibble: 3 x 2
  geno  median
  <chr>  <dbl>
1 A/A     31.2
2 A/G     25.1
3 G/G     20.1
```

The median ORMDL3 gene expression level for each genotype is as follows: 31.2 for A/A, 25.1 for A/G, and 20.1 for G/G.

Method 2: We can also generate and examine a base R boxplot of the data.

```
# Store boxplot output of expression level vs. genotype to an object
expr.box <- boxplot(exp ~ geno, data = expr)
```




```r
# Examine stat values from boxplot object
expr.box$stats
```

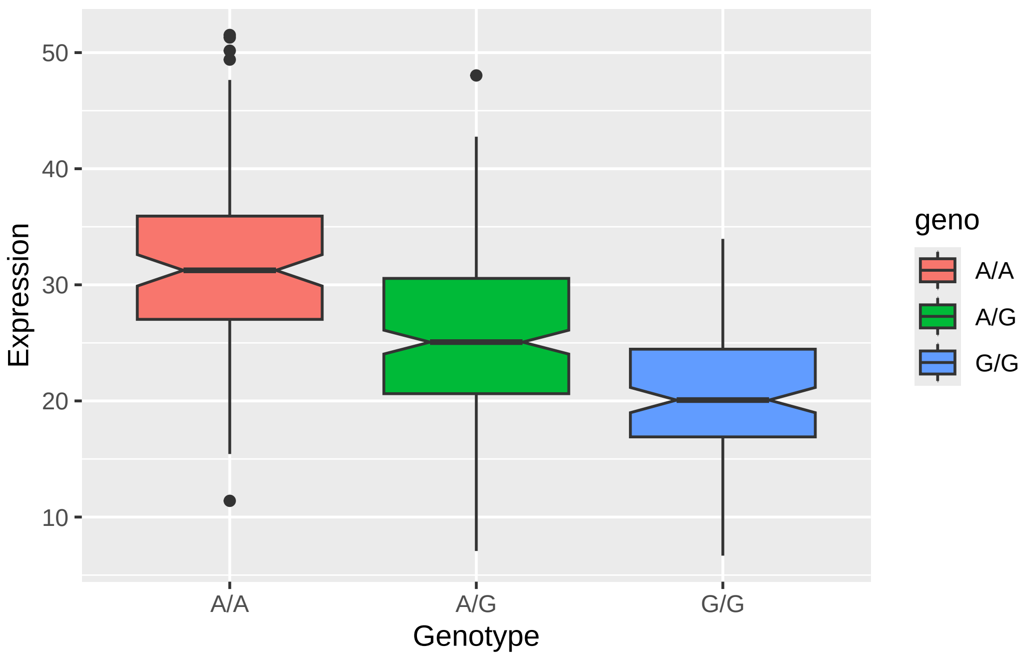
```
         [,1]     [,2]     [,3]
[1,] 15.42908  7.07505  6.67482
[2,] 26.95022 20.62572 16.90256
[3,] 31.24847 25.06486 20.07363
[4,] 35.95503 30.55183 24.45672
[5,] 49.39612 42.75662 33.95602
```

By saving the boxplot output as an object and accessing the object's `stats` matrix, we can examine the lower whisker, 1st quartile, median, 3rd quartile, and upper whisker expression levels (rows 1-5, respectively) for genotypes A/A, A/G, and G/G (columns 1-3, respectively). The median expression levels (`expr.box$stats[3,]`) for A/A, A/G, and G/G are 31.25, 25.06, and 20.07, respectively.

> Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

Let's make a ggplot2 boxplot of expression level vs. genotype:

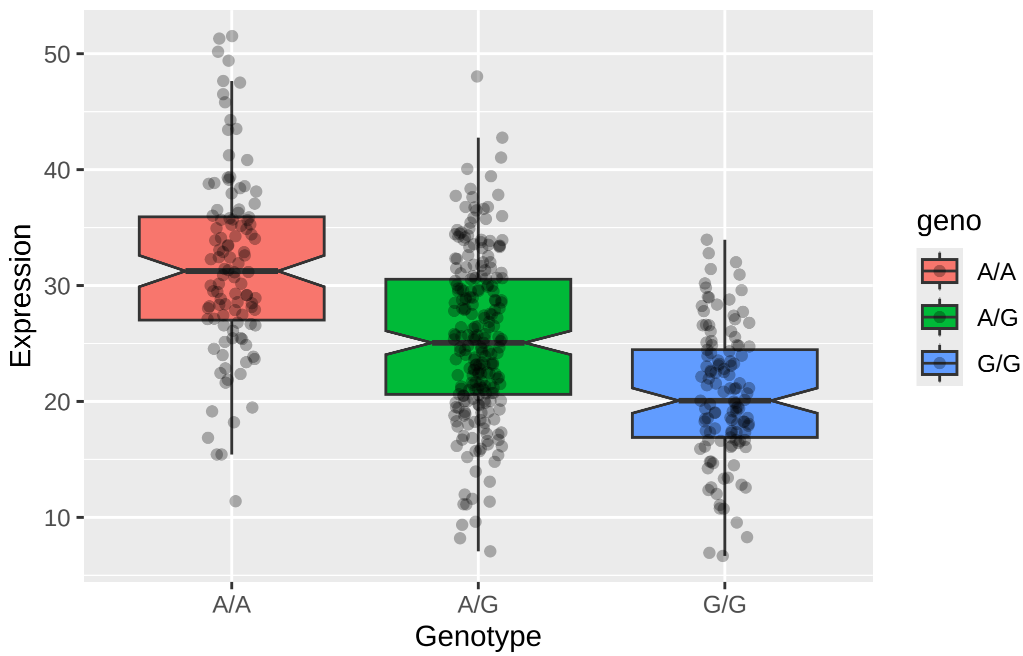
```r
library(ggplot2)

ggplot(expr) +
  aes(x = geno, y = exp, fill = geno) +
  geom_boxplot(notch = TRUE) +
  labs(x = "Genotype", y = "Expression")
```



From the boxplot, it seems the 3rd quartile / 75th percentile expression level of G/G individuals sampled is lower than the 1st quartile / 25th percentile expression level of A/A individuals sampled. Thus, it appears the G/G genotype is associated with lower expression of ORMDL3 than the A/A genotype, and the SNP *does* affect the expression of ORMDL3.

Note: Extra boxplot, with raw data points displayed over the boxplot using the additional, `geom_jitter()` layer:

```r
ggplot(expr) +
  aes(x = geno, y = exp, fill = geno) +
  geom_boxplot(notch = TRUE, outliers = FALSE) +
  geom_jitter(alpha = 0.3, width = 0.1) +
  labs(x = "Genotype", y = "Expression")
```



Expression
50
40
30
20
10
A/A
A/G
G/G
Genotype
geno
A/A
A/G
G/G